
Samsca® Tablets 15 mg

(Tolvaptan)

FULL PRESCRIBING INFORMATION

**WARNING: (A) INITIATE AND RE-INITIATE IN A HOSPITAL AND MONITOR
SERUM SODIUM**

**(B) NOT FOR USE FOR AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY
DISEASE (ADPKD)**

(A) Initiate and re-initiate in a hospital and monitor serum sodium

SAMSCA should be initiated and re-initiated in patients only in a hospital where serum sodium can be monitored closely.

Too rapid correction of hyponatremia (e.g., >12 mEq/L/24 hours) can cause osmotic demyelination resulting in dysarthria, mutism, dysphagia, lethargy, affective changes, spastic quadriparesis, seizures, coma and death. In susceptible patients including those with severe malnutrition, alcoholism or advanced liver disease, slower rates of correction may be advisable.

For volume overload in heart failure, administration of SAMSCA should be initiated or re-initiated only under hospitalization, as occurrence of consciousness disturbance has been reported as a result of dehydration and hypernatremia associated with rapid water diuresis and rapid increase in serum sodium concentration may induce osmotic demyelination syndrome (ODS). On the first day of treatment initiation or re-initiation, serum sodium concentration should be frequently monitored.

(B) Not for use for autosomal dominant polycystic kidney disease (ADPKD)

Because of the risk of hepatotoxicity, SAMSCA should not be used for ADPKD [see *Contraindications* (4.1)].

1 INDICATIONS AND USAGE

1.1 Hyponatremia

The treatment of clinically significant hypervolemic and euvolemic hyponatremia [serum sodium < 125 mEq/L or hyponatremia that is symptomatic and has resisted correction with fluid restriction], including patients with heart failure, and Syndrome of Inappropriate Antidiuretic Hormone (SIADH).

Important Limitations

Patients requiring intervention to raise serum sodium urgently to prevent or to treat serious neurological symptoms should not be treated with SAMSCA.

It has not been established that raising serum sodium with SAMSCA provides a symptomatic benefit to patients.

1.2 Adjunct Treatment of Volume Overload in Heart Failure

Volume overload in heart failure when adequate response is not obtained with other diuretics (e.g., loop diuretics).

Important Limitations

SAMSCA should be used in combination with other diuretics, such as loop diuretics, thiazides, and aldosterone antagonists.

2 DOSAGE AND ADMINISTRATION

2.1 Usual Dosage in Adults

Patients should be in a hospital for initiation and re-initiation of therapy to evaluate the therapeutic response and because too rapid correction of hyponatremia can cause osmotic demyelination resulting in dysarthria, mutism, dysphagia, lethargy, affective changes, spastic quadriparesis, seizures, coma and death.

During initiation and titration, frequently monitor for change in serum electrolytes and volume. Fluid restriction during the first 24 hours of therapy with SAMSCA should generally be avoided. Patients receiving SAMSCA should continue ingestion of fluid in response to thirst [*see Warnings and Precautions (5.1)*].

In order to avoid nocturnal urination, SAMSCA is recommended to be administered in the morning and swallowed whole with water regardless of meal.

Hyponatremia

The usual starting dose for SAMSCA is 15 mg administered once daily without regard to meals. Increase the dose to 30 mg once daily, after at least 24 hours, to a maximum of 60 mg once daily, as needed to achieve the desired level of serum sodium. Do not administer SAMSCA for more than 30 days to minimize the risk of liver injury for the treatment of hyponatremia.

Adjunct treatment of volume overload in heart failure

SAMSCA should be used in combination with other diuretics, such as loop diuretics, thiazides, and aldosterone antagonists, since SAMSCA increases aquaresis but not natriuresis. There is no clinical experience of co-administration of SAMSCA with human atrial natriuretic peptide (hANP).

The usual adult dose is 15 mg once daily but it is recommended to start from 7.5 mg/day for patients whose serum sodium is less than 125 mEq/L, or for patients in whom rapid volume decrease is undesirable, or for elderly patients or patients with serum sodium concentration higher than 140 mEq/L.

When conditions or symptoms due to volume overload are resolved, administration of SAMSCA should be discontinued. The efficacy of SAMSCA for maintenance treatment following resolution has not been evaluated.

Administration of SAMSCA should not be continued after body weight has returned to the target level (body weight at which volume overload is appropriately controlled).

Administration of SAMSCA should not be continued if volume overload or body fluid retention is not improved.

2.2 Drug Withdrawal

Following discontinuation from SAMSCA, patients should be advised to resume fluid restriction and should be monitored for changes in serum sodium and volume status.

2.3 Co-Administration with CYP 3A Inhibitors, CYP 3A Inducers and P-gp Inhibitors

CYP 3A Inhibitors

SAMSCA is metabolized by CYP 3A, and use with strong CYP 3A inhibitors causes a marked (5-fold) increase in exposure [see *Contraindications* (4.5)]. The effect of moderate CYP 3A inhibitors on SAMSCA exposure has not been assessed. Avoid co-administration of SAMSCA and moderate CYP 3A inhibitors [see *Warnings and Precautions* (5.6), *Drug Interactions* (7.1)].

CYP 3A Inducers

Co-administration of SAMSCA with potent CYP 3A inducers (e.g., rifampin) reduces SAMSCA plasma concentrations by 85%. Therefore, the expected clinical effects of SAMSCA may not be observed at the recommended dose. Patient response should be monitored and the dose adjusted accordingly [see *Warnings and Precautions* (5.6), *Drug Interactions* (7.1)].

P-gp Inhibitors

SAMSCA is a substrate of P-gp. Co-administration of SAMSCA with inhibitors of P-gp (e.g., cyclosporine) may necessitate a decrease in SAMSCA dose [see *Warnings and Precautions* (5.6), *Drug Interactions* (7.1)].

3 DOSAGE FORM AND STRENGTH

SAMSCA is available as 15 mg tablets [see *How Supplied/Storage and Handling* (14)].

4 CONTRAINDICATIONS

SAMSCA is contraindicated in the following conditions:

4.1 Use in patients with Autosomal Dominant Polycystic Kidney Disease (ADPKD)

Samsca can cause serious and potentially fatal liver injury. SAMSCA should not be prescribed or used outside of the approved indications.

4.2 Urgent need to raise serum sodium acutely.

SAMSCA has not been studied in a setting of urgent need to raise serum sodium acutely.

4.3 Inability of the patient to sense or appropriately respond to thirst.

Patients who are unable to auto-regulate fluid balance are at substantially increased risk of incurring an overly rapid correction of serum sodium, hyponatremia and hypovolemia.

4.4 Hypovolemic hyponatremia

Risks associated with worsening hypovolemia, including complications such as

hypotension and renal failure, outweigh possible benefits.

4.5 Concomitant use of strong CYP3A inhibitors

Ketoconazole 200 mg administered with SAMSCA increased SAMSCA exposure by 5-fold. Larger doses would be expected to produce larger increases in SAMSCA exposure. There is no adequate experience to define the dose adjustment that would be needed to allow safe use of SAMSCA with strong CYP 3A inhibitors such as clarithromycin, ketoconazole, itraconazole, ritonavir, indinavir, nelfinavir, saquinavir, nefazodone, and telithromycin.

4.6 Anuric patients

In patients unable to make urine, no clinical benefit can be expected.

4.7 Hypersensitivity

SAMSCA is contraindicated in patients with hypersensitivity (e.g. anaphylactic shock, rash generalized) to tolvaptan or any component of the product [see Adverse Reactions (6.2)].

4.8 Hyponatremia patients

5 WARNINGS AND PRECAUTIONS

5.1 Too rapid correction of serum sodium can cause serious neurologic sequelae (see BOXED WARNING)

Osmotic demyelination syndrome is a risk associated with too rapid correction of hyponatremia (e.g., > 12 mEq/L/24 hours). Osmotic demyelination results in dysarthria, mutism, dysphagia, lethargy, affective changes or spastic quadriparesis, seizures, coma and death. In susceptible patients, including those with hypoxia, severe malnutrition, alcoholism or advanced liver disease, slower rates of correction may be advisable. In controlled clinical trials in which SAMSCA was administered in titrated doses starting at 15 mg once daily, 7% of SAMSCA treated subjects with a serum sodium < 130 mEq/L had an increase in serum sodium greater than 8 mEq/L at approximately 8 hours and 2% had an increase greater than 12 mEq/L at 24 hours.

Approximately 1% of placebo-treated subjects with a serum sodium < 130 mEq/L had a rise greater than 8 mEq/L at 8 hours and no patient had a rise greater than 12 mEq/L/24 hours.

Osmotic demyelination syndrome has been reported in association with SAMSCA therapy [see *Adverse Reactions (6.2)*]. Patients treated with SAMSCA should be monitored to assess serum sodium concentrations and neurologic status, especially during initiation and after titration. Subjects with SIADH or very low baseline serum sodium concentrations may be at greater risk for too-rapid correction of serum sodium. In patients receiving SAMSCA who develop too rapid a rise in serum sodium (>12 mEq/L/24 hours), discontinue or interrupt treatment with SAMSCA and consider administration of hypotonic fluid. Fluid restriction during the first 24 hours of therapy with SAMSCA may increase the likelihood of overly-rapid correction of serum sodium, and should generally be avoided. Co-administration of diuretics also increases the risk of too rapid correction of serum sodium and such patients should undergo close monitoring of serum sodium.

5.2 Hyponatremia (see BOXED WARNING)

Hemoconcentration associated with rapid diuresis may induce hyponatremia, possibly accompanied by consciousness disturbance. In patients being treated with SAMSCA, fluid intake, urine volume, serum sodium level, and the occurrence of clinical symptoms such as thirst and dehydration should be carefully monitored. If clinical symptoms such as persistent thirst and dehydration are observed, SAMSCA should be discontinued or the dosage reduced and appropriate measures such as fluid replenishment, including fluid therapy, should be taken in accordance with the symptoms. If serum sodium level increases above the normal range, administration of SAMSCA should be discontinued immediately and appropriate measures such as fluid replenishment, including fluid therapy, should be taken.

5.3 Liver Injury

SAMSCA can cause serious and potentially fatal liver injury. In placebo-controlled studies and an open-label extension study of chronically administered tolvaptan in patients with ADPKD, cases of serious liver injury attributed to tolvaptan, generally occurring during the first 18 months of therapy, were observed. In postmarketing experience with tolvaptan in ADPKD, acute liver failure requiring liver transplantation has been reported. SAMSCA should not be used to treat ADPKD [see *Contraindications (4.1)*].

Patients with symptoms that may indicate liver injury, including fatigue, anorexia, right upper abdominal discomfort, dark urine or jaundice should discontinue treatment with SAMSCA.

Limit duration of therapy with SAMSCA to 30 days for treatment of hyponatremia. Avoid use in patients with underlying liver disease, including cirrhosis, because the ability to recover from liver injury may be impaired [see *Adverse Reactions (6.1)*].

For the treatment of volume overload in heart failure, liver function tests should be performed prior to initiation of SAMSCA and frequently at least for the first 2 weeks of treatment. For long term use (i.e. more than 30 days), blood testing for hepatic

transaminases and bilirubin is required prior to initiation of SAMSCA, and continuing monthly for 18 months and at regular intervals (e.g., 3-6 month) thereafter.

5.4 Dehydration and Hypovolemia

SAMSCA therapy induces copious aquaresis, which is normally partially offset by fluid intake. Dehydration and hypovolemia can occur, especially in potentially volume-depleted patients receiving diuretics or those who are fluid restricted. In multiple-dose, placebo-controlled trials in which 607 hyponatremic patients were treated with tolvaptan, the incidence of dehydration was 3.3% for tolvaptan and 1.5% for placebo-treated patients.

As the aquaretic effect of SAMSCA is most potent within 24 hours after the initial administration, serum sodium concentration should be measured at least at 4 to 6 hours and 8 to 12 hours after administration on the first day of administration. From the second to around the seventh day of administration, serum sodium concentration should be measured every day, and if administration is continued further, serum sodium concentration should be measured at appropriate intervals.

In patients receiving SAMSCA who develop medically significant signs or symptoms of hypovolemia, interrupt or discontinue SAMSCA therapy and provide supportive care with careful management of vital signs, fluid balance and electrolytes. Fluid restriction during therapy with SAMSCA may increase the risk of dehydration and hypovolemia. Patients receiving SAMSCA should continue ingestion of fluid in response to thirst. If sensation of thirst persists, dose reduction should be considered.

5.5 Co-administration with hypertonic saline

Concomitant use with hypertonic saline is not recommended

5.6 Drug Interactions

Other drugs affecting exposure to SAMSCA

CYP 3A Inhibitors

SAMSCA is a substrate of CYP 3A. CYP 3A inhibitors can lead to a marked increase in SAMSCA concentrations [see [Dosage and Administration \(2.3\)](#), [Drug Interactions \(7.1\)](#)]. Do not use SAMSCA with strong inhibitors of CYP3A and avoid concomitant use with moderate CYP 3A inhibitors [see [Contraindications \(4.5\)](#)].

CYP 3A Inducers

Avoid co-administration of CYP 3A inducers (e.g., rifampin, rifabutin, rifapentin, barbiturates, phenytoin, carbamazepine, St. John's Wort) with SAMSCA, as this can

lead to a reduction in the plasma concentration of SAMSCA and decreased effectiveness of SAMSCA treatment. If co-administered with CYP 3A inducers, the dose of SAMSCA may need to be increased [see Dosage and Administration (2.3), Drug Interactions (7.1)].

P-gp Inhibitors

The dose of SAMSCA may have to be reduced when SAMSCA is co-administered with P-gp inhibitors, e.g., cyclosporine [see Dosage and Administration (2.3), Drug Interactions (7.1)].

5.7 Hyperkalemia or Drugs that Increase Serum Potassium

Treatment with SAMSCA is associated with an acute reduction of the extracellular fluid volume which could result in increased serum potassium and may lead to ventricular fibrillation and ventricular tachycardia. Serum potassium levels should be monitored after initiation of SAMSCA treatment in patients with a serum potassium > 5 mEq/L as well as those who are receiving drugs known to increase serum potassium levels.

5.8 Effects on Ability to Drive and Use Machines

There are no controlled trials of the effects of tolvaptan on driving performance. When driving vehicles or using machines it should be taken into account that occasionally dizziness, asthenia or syncope may occur.

5.9 Urinary Outflow Obstruction

Urinary output must be secured. Patients with partial obstruction of urinary outflow, for example patients with prostatic hypertrophy or impairment of micturition, have an increased risk of developing acute retention.

5.10 Patients with serious coronary artery disease or cerebrovascular disease and elderly patients.

Rapid volume decrease or hemoconcentration associated with rapid diuresis may induce thromboembolism.

6 ADVERSE REACTIONS

6.1 Clinical Trials Experience

6.1.1 Hyponatremia

Because clinical trials are conducted under widely varying conditions, adverse reactions rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in practice. The adverse event information from clinical trials does, however, provide a basis for identifying the adverse events that appear to be related to drug use and for approximating rates.

In multiple-dose, placebo-controlled trials, 607 hyponatremic patients (serum sodium < 135 mEq/L) were treated with SAMSCA. The mean age of these patients was 62 years; 70% of patients were male and 82% were Caucasian. One hundred eighty-nine (189) SAMSCA-treated patients had a serum sodium < 130 mEq/L, and 52 patients had a serum sodium < 125 mEq/L. Hyponatremia was attributed to cirrhosis in 17% of patients, heart failure in 68% and SIADH/other in 16%. Of these patients, 223 were treated with the recommended dose titration (15 mg titrated to 60 mg as needed to raise serum sodium).

Overall, over 4,000 patients have been treated with oral doses of SAMSCA in open-label or placebo-controlled clinical trials. Approximately 650 of these patients had hyponatremia; approximately 219 of these hyponatremic patients were treated with SAMSCA for 6 months or more.

The most common adverse reactions (incidence $\geq$ 5% more than placebo) seen in two 30-day, double-blind, placebo-controlled hyponatremia trials in which SAMSCA was administered in titrated doses (15 mg to 60 mg once daily) were thirst, dry mouth, asthenia, constipation, pollakiuria or polyuria and hyperglycemia. In these trials, 10% (23/223) of SAMSCA-treated patients discontinued treatment because of an adverse event, compared to 12% (26/220) of placebo-treated patients; no adverse reaction resulting in discontinuation of trial medication occurred at an incidence of > 1% in SAMSCA-treated patients.

Table 1 lists the adverse reactions reported in SAMSCA-treated patients with hyponatremia (serum sodium < 135 mEq/L) and at a rate at least 2% greater than placebo-treated patients in two 30-day, double-blind, placebo-controlled trials. In these studies, 223 patients were exposed to SAMSCA (starting dose 15 mg, titrated to 30 and 60 mg as needed to raise serum sodium). Adverse events resulting in death in these trials were 6% in SAMSCA-treated-patients and 6% in placebo-treated patients.

Table 1. Adverse Reactions (> 2% more than placebo) in SAMSCA-Treated Patients in Double-Blind, Placebo-Controlled Hyponatremia Trials

System Organ Class MedDRA Preferred Term	SAMSCA 15 mg/day-60 mg/day (N = 223) n (%)	Placebo (N = 220) n (%)
Gastrointestinal Disorders		
Dry mouth	28 (13)	9 (4)
Constipation	16 (7)	4 (2)
General Disorders and Administration Site Conditions		
Thirst ^a	35 (16)	11 (5)
Asthenia	19 (9)	9 (4)
Pyrexia	9 (4)	2 (1)
Metabolism and Nutrition Disorders		
Hyperglycemia ^b	14 (6)	2 (1)
Anorexia ^c	8 (4)	2 (1)
Renal and Urinary Disorders		
Pollakiuria or polyuria ^d	25 (11)	7 (3)

The following terms are subsumed under the referenced ADR in Table 1:

^a polydipsia; ^b diabetes mellitus; ^c decreased appetite; ^d urine output increased, micturition urgency, nocturia

In a subgroup of patients with hyponatremia (N = 475, serum sodium < 135 mEq/L) enrolled in a double-blind, placebo-controlled trial (mean duration of treatment was 9 months) of patients with worsening heart failure, the following adverse reactions occurred in SAMSCA-treated patients at a rate at least 2% greater than placebo: mortality (42% SAMSCA, 38% placebo), nausea (21% SAMSCA, 16% placebo), thirst (12% SAMSCA, 2% placebo), dry mouth (7% SAMSCA, 2% placebo) and polyuria or pollakiuria (4% SAMSCA, 1% placebo).

Gastrointestinal bleeding in patients with cirrhosis

In patients with cirrhosis treated with tolvaptan in the hyponatremia trials, gastrointestinal bleeding was reported in 6 out of 63 (10%) tolvaptan-treated patients and 1 out of 57 (2%) placebo treated patients.

The following adverse reactions occurred in < 2% of hyponatremic patients treated with SAMSCA and at a rate greater than placebo in double-blind placebo-controlled trials (N = 607 SAMSCA; N = 518 placebo) or in < 2% of patients in an uncontrolled trial of patients with hyponatremia (N = 111) and are not mentioned elsewhere in the label.

Blood and Lymphatic System Disorders: Disseminated intravascular coagulation

Cardiac Disorders: Intracardiac thrombus, ventricular fibrillation

Investigations: Prothrombin time prolonged

Gastrointestinal Disorders: Ischemic colitis

Metabolism and Nutrition Disorders: Diabetic ketoacidosis

Musculoskeletal and Connective Tissue Disorders: Rhabdomyolysis

Nervous System: Cerebrovascular accident

Renal and Urinary Disorders: Urethral hemorrhage

Reproductive System and Breast Disorders (female): Vaginal hemorrhage

Respiratory, Thoracic, and Mediastinal Disorders: Pulmonary embolism, respiratory failure

Vascular disorder: Deep vein thrombosis

6.1.2 Adjunct treatment of volume overload in heart failure (Japanese data)

Table 2 shows the adverse reactions reported in Japanese clinical trials of tolvaptan in patients with volume overload in heart failure (N= 213). Included are adverse events that were reported in $\geq 2\%$ of patients who received any oral tolvaptan dose and assessed as reasonably associated with tolvaptan use.

Table 2. Adverse Reactions in SAMSCA-Treated Patients in Japanese Clinical Trials for Volume Overload in Heart Failure

Gastrointestinal Disorders	
Very common ($\geq 10\%$)	Constipation
Common ($\geq 2\%$ and $< 10\%$)	Diarrhea
General Disorders and Administration Site Conditions	
Very common ($\geq 10\%$)	Thirst
Common ($\geq 2\%$ and $< 10\%$)	Malaise
Investigations	
Very common ($\geq 10\%$)	Blood creatinine increased, Blood urea increased, Blood uric acid increased
Common ($\geq 2\%$ and $< 10\%$)	Blood glucose increased, Blood potassium increased, Blood pressure decreased, Blood urine present
Nervous system disorders	
Common ($\geq 2\%$ and $< 10\%$)	Dizziness ^a , Headache
Renal and Urinary Disorders	

Common ($\geq 2\%$ and $< 10\%$)	Pollakiuria
------------------------------------	-------------

The following term is subsumed under the referenced ADR in Table 2: ^a Dizziness postural

6.2 Postmarketing Experience

The following adverse reactions have been identified during post-approval use of SAMSCA. Because these reactions are reported voluntarily from a population of an unknown size, it is not always possible to reliably estimate their frequency or establish a causal relationship to drug exposure.

Neurologic: Osmotic demyelination syndrome

Investigations: Hyponatremia

Removal of excess free body water increases serum osmolality and serum sodium concentrations. All patients treated with SAMSCA, especially those whose serum sodium levels become normal, should continue to be monitored to ensure serum sodium remains within normal limits. If hyponatremia is observed, management may include dose decreases or interruption of SAMSCA treatment, combined with modification of free-water intake or infusion. During clinical trials of hyponatremia patients, hyponatremia was reported as an adverse event in 0.7% of patients receiving SAMSCA vs. 0.6% of patients receiving placebo; analysis of laboratory values demonstrated an incidence of hyponatremia of 1.7% in patients receiving SAMSCA vs. 0.8% in patients receiving placebo.

Immune system disorders: Hypersensitivity reactions including anaphylactic shock and rash generalized [*see Contraindications (4.7)*].

7 DRUG INTERACTIONS

7.1 Effects of drugs on SAMSCA

Ketoconazole and Other Strong CYP3A Inhibitors

SAMSCA is metabolized primarily by CYP 3A. Ketoconazole is a strong inhibitor of CYP 3A and also an inhibitor of P-gp. Co-administration of SAMSCA and ketoconazole 200 mg daily results in a 5-fold increase in exposure to SAMSCA. Co-administration of SAMSCA with 400 mg ketoconazole daily or with other strong CYP 3A inhibitors (e.g., clarithromycin, itraconazole, telithromycin, saquinavir, nelfinavir, ritonavir and nefazodone) at the highest labeled dose would be expected to cause an even greater increase in SAMSCA exposure. Thus, SAMSCA and strong CYP 3A inhibitors should not be co-administered [*see Dosage and Administration (2.3) and Contraindications (4.5)*].

Moderate CYP3A Inhibitors

The impact of moderate CYP 3A inhibitors (e.g., erythromycin, fluconazole, aprepitant, diltiazem and verapamil) on the exposure to co-administered SAMSCA has not been

assessed. A substantial increase in the exposure to SAMSCA would be expected when SAMSCA is co-administered with moderate CYP 3A inhibitors. Co-administration of moderate CYP3A inhibitors with SAMSCA should generally be avoided [see Dosage and Administration (2.3) and Warnings & Precautions (5.6)].

Grapefruit Juice

Co-administration of grapefruit juice and SAMSCA results in a 1.9-fold increase in exposure to SAMSCA. Patients taking SAMSCA should avoid ingesting grapefruit juice [see Dosage and Administration (2.3) and Warnings & Precautions (5.6)].

P-gp Inhibitors

Reduction in the dose of SAMSCA may be required in patients concomitantly treated with P-gp inhibitors, such as e.g., cyclosporine, based on clinical response [see Dosage and Administration (2.3) and Warnings & Precautions (5.6)].

Rifampin and Other CYP 3A Inducers

Rifampin is an inducer of CYP 3A and P-gp. Co-administration of rifampin and SAMSCA reduces exposure to SAMSCA by 85%. Therefore, the expected clinical effects of SAMSCA in the presence of rifampin and other inducers (e.g., rifabutin, rifapentin, barbiturates, phenytoin, carbamazepine and St. John's Wort) may not be observed at the usual dose levels of SAMSCA. The dose of SAMSCA may have to be increased [see Dosage and Administration (2.3) and Warnings & Precautions (5.6)].

Lovastatin, Digoxin, Furosemide, and Hydrochlorothiazide

Co-administration of lovastatin, digoxin, furosemide, and hydrochlorothiazide with SAMSCA has no clinically relevant impact on the exposure to SAMSCA.

7.2 Effects of SAMSCA on Other Drugs

Digoxin

Digoxin is a P-gp substrate and SAMSCA is a P-gp inhibitor. Co-administration of SAMSCA with digoxin increased digoxin AUC by 20% and C_{max} by 30%.

Warfarin, Amiodarone, Furosemide, and Hydrochlorothiazide

Co-administration of SAMSCA does not appear to alter the pharmacokinetics of warfarin, furosemide, hydrochlorothiazide, or amiodarone (or its active metabolite, desethylamiodarone) to a clinically significant degree.

Lovastatin

SAMSCA is a weak inhibitor of CYP 3A. Co-administration of lovastatin and SAMSCA

increases the exposure to lovastatin and its active metabolite lovastatin- β hydroxyacid by factors of 1.4 and 1.3, respectively. This is not a clinically relevant change.

Pharmacodynamic Interactions

SAMSCA produces a greater 24 hour urine volume and 24 hour excretion rate than does furosemide or hydrochlorothiazide. Concomitant administration of SAMSCA with furosemide or hydrochlorothiazide results in a 24 hour urine volume and 24 hour excretion rate that is similar to the rate after SAMSCA administration alone.

Although specific interaction studies were not performed, in clinical studies SAMSCA was used concomitantly with beta-blockers, angiotensin receptor blockers, angiotensin converting enzyme inhibitors and potassium sparing diuretics. Adverse reactions of hyperkalemia were approximately 1-2% higher when SAMSCA was administered with angiotensin receptor blockers, angiotensin converting enzyme inhibitors and potassium sparing diuretics compared to administration of these medications with placebo. Serum potassium levels should be monitored during concomitant drug therapy.

As a V2 receptor antagonist, SAMSCA may interfere with the V2 agonist activity of desmopressin (dDAVP). In a male subject with mild Von Willebrand (vW) disease, intravenous infusion of dDAVP 2 hours after administration of oral SAMSCA did not produce the expected increases in vW Factor Antigen or Factor VIII activity. It is not recommended to administer SAMSCA with a V2 agonist.

8 USE IN SPECIFIC POPULATIONS

There is no need to adjust dose based on age, gender, race, or cardiac function [see Clinical Pharmacology (11.3)].

8.1 Pregnancy

Pregnancy Category C

There are no adequate and well controlled studies of SAMSCA use in pregnant women. In animal studies, cleft palate, brachymelia, microphthalmia, skeletal malformations, decreased fetal weight, delayed fetal ossification, and embryo-fetal death occurred. SAMSCA should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

In embryo-fetal development studies, pregnant rats and rabbits received oral SAMSCA during organogenesis. Rats received 2 to 162 times the maximum recommended human dose (MRHD) of SAMSCA (on a body surface area basis). Reduced fetal weights and delayed fetal ossification occurred at 162 times the MRHD. Signs of maternal toxicity (reduction in body weight gain and food consumption) occurred at 16 and 162 times the MRHD. When pregnant rabbits received oral SAMSCA at 32 to 324

times the MRHD (on a body surface area basis), there were reductions in maternal body weight gain and food consumption at all doses, and increased abortions at the mid and high doses (about 97 and 324 times the MRHD). At 324 times the MRHD, there were increased rates of embryo-fetal death, fetal microphthalmia, open eyelids, cleft palate, brachymelia and skeletal malformations [see Nonclinical Toxicology (12.2)].

8.2 Labor and Delivery

The effect of SAMSCA on labor and delivery in humans is unknown.

8.3 Nursing Mothers

It is not known whether SAMSCA is excreted into human milk. SAMSCA is excreted into the milk of lactating rats. Because many drugs are excreted into human milk and because of the potential for serious adverse reactions in nursing infants from SAMSCA, a decision should be made to discontinue nursing or SAMSCA, taking into consideration the importance of SAMSCA to the mother.

8.4 Pediatric Use

Safety and effectiveness of SAMSCA in pediatric patients have not been established.

8.5 Geriatric Use

A rapid decrease in circulating plasma volume or hemoconcentration associated with rapid diuresis may induce thromboembolism. SAMSCA should be administered with care and the patient's condition should be closely monitored. Elderly patients generally have reduced physiological function and are known to be susceptible to dehydration.

Hyponatremia

Of the total number of hyponatremic subjects treated with SAMSCA in clinical studies, 42% were 65 and over, while 19% were 75 and over. No overall differences in safety or effectiveness were observed between these subjects and younger subjects, and other reported clinical experience has not identified differences in responses between the elderly and younger patients, but greater sensitivity of some older individuals cannot be ruled out. Increasing age has no effect on SAMSCA plasma concentrations.

Adjunct treatment of volume overload in heart failure

Initiation of SAMSCA at half dose (7.5mg) is recommended in elderly patients with volume overload due to heart failure, because elderly patients were found to be at risk for hypernatremia.

8.6 Use in Patients with Hepatic Impairment

Moderate and severe hepatic impairment do not affect exposure to tolvaptan to a clinically relevant extent. Avoid use of tolvaptan in patients with underlying liver disease.

8.7 Use in Patients with Renal Impairment

No dose adjustment is necessary based on renal function. There are no clinical trial data in patients with CrCl <10 mL/min, and, because drug effects on serum sodium levels are likely lost at very low levels of renal function, use in patients with a CrCl <10 mL/min is not recommended. No benefit can be expected in patients who are anuric [see *Contraindications (4.6)*, *Clinical Pharmacology (11.3)*].

8.8 Use in Patients with Congestive Heart Failure

The exposure to SAMSCA in patients with congestive heart failure is not clinically relevantly increased. No dose adjustment is necessary.

9 OVERDOSE

Single oral doses up to 480 mg and multiple doses up to 300 mg once daily for 5 days have been well tolerated in studies in healthy subjects. There is no specific antidote for SAMSCA intoxication. The signs and symptoms of an acute overdose can be anticipated to be those of excessive pharmacologic effect: a rise in serum sodium concentration, polyuria, thirst, and dehydration/hypovolemia.

The oral LD₅₀ of SAMSCA in rats and dogs is > 2000 mg/kg. No mortality was observed in rats or dogs following single oral doses of 2000 mg/kg (maximum feasible dose). A single oral dose of 2000 mg/kg was lethal in mice, and symptoms of toxicity in affected mice included decreased locomotor activity, staggering gait, tremor and hypothermia.

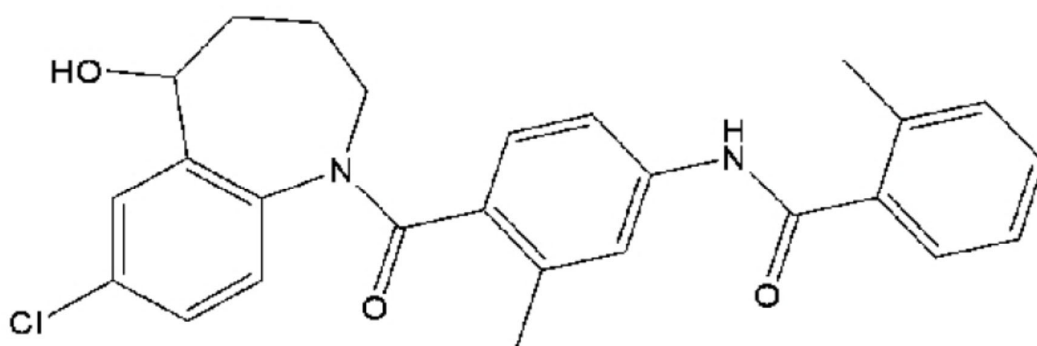
If overdose occurs, estimation of the severity of poisoning is an important first step. A thorough history and details of overdose should be obtained, and a physical examination should be performed. The possibility of multiple drug involvement should be considered.

Treatment should involve symptomatic and supportive care, with respiratory, ECG and blood pressure monitoring and water/electrolyte supplements as needed. A profuse and prolonged aquaresis should be anticipated, which, if not matched by oral fluid ingestion, should be replaced with intravenous hypotonic fluids, while closely monitoring electrolytes and fluid balance.

ECG monitoring should begin immediately and continue until ECG parameters are within normal ranges. Dialysis may not be effective in removing SAMSCA because of its high binding affinity for human plasma protein (> 98 %). Close medical supervision and monitoring should continue until the patient recovers.

10 DESCRIPTION

Tolvaptan is (±)-4'-[(7-chloro-2,3,4,5-tetrahydro-5-hydroxy-1*H*-1-benzazepin-1-yl) carbonyl]-*o*-tolu-*m*-toluidide. The empirical formula is C₂₆H₂₅ClN₂O₃. Molecular weight is 448.94. The chemical structure is:



SAMSCA tablet for oral use contains 15 mg of tolvaptan. Inactive ingredients include:

FD&C Blue No. 2 Aluminum Lake, Lactose monohydrate (Cow, milk), Microcrystalline cellulose, Magnesium stearate, Corn starch, Low Substituted Hydroxypropylcellulose, Hydroxypropylcellulose

11 CLINICAL PHARMACOLOGY

WHO ATC Code: C03XA01

11.1 Mechanism of action

SAMSCA is a selective vasopressin V₂-receptor antagonist with an affinity for the V₂-receptor that is 1.8 times that of native arginine vasopressin (AVP). SAMSCA affinity for the V₂-receptor is 29 times greater than for the V_{1a}-receptor. When taken orally, 15 to 60 mg doses of SAMSCA antagonize the effect of vasopressin and cause an increase in urine water excretion that results in an increase in free water clearance (aquaresis), a decrease in urine osmolality, and a resulting increase in serum sodium concentrations. Urinary excretion of sodium and potassium and plasma potassium concentrations are not significantly changed. SAMSCA metabolites have no or weak antagonist activity for human V₂-receptors compared with SAMSCA.

Plasma concentrations of native AVP may increase (avg. 2-9 pg/mL) with SAMSCA administration.

11.2 Pharmacodynamics

In healthy subjects receiving a single dose of SAMSCA 60 mg, the onset of the aquaretic and sodium increasing effects occurs within 2 to 4 hours post-dose. A peak effect of about a 6 mEq increase in serum sodium and about 9 mL/min increase in urine excretion rate is observed between 4 and 8 hours post-dose; thus, the pharmacological activity lags behind the plasma concentrations of SAMSCA. About 60% of the peak effect on serum sodium is sustained at 24 hours post-dose, but the urinary excretion rate is no longer elevated by this time. Doses above 60 mg SAMSCA do not increase aquaresis or serum sodium further. The effects of SAMSCA in the recommended dose range of 15 to 60 mg once daily appear to be limited to aquaresis and the resulting increase in sodium concentration.

In a parallel-arm, double-blind (for SAMSCA and placebo), placebo-and positive-controlled, multiple dose study of the effect of SAMSCA on the QTc interval, 172 healthy subjects were randomized to SAMSCA 30 mg, SAMSCA 300 mg, placebo, or moxifloxacin 400 mg once daily. At both the 30 mg and 300 mg doses, no significant effect of administering SAMSCA on the QTc interval was detected on Day 1 and Day 5. At the 300 mg dose, peak SAMSCA plasma concentrations were approximately 4-fold higher than the peak concentrations following a 30 mg dose. Moxifloxacin increased the QT interval by 12 ms at 2 hours after dosing on Day 1 and 17 ms at 1 hour after dosing on Day 5, indicating that the study was adequately designed and conducted to detect SAMSCA's effect on the QT interval, had an effect been present.

11.3 Pharmacokinetics

In healthy subjects the pharmacokinetics of tolvaptan after single doses of up to 480 mg and multiple doses up to 300 mg once daily have been examined. Area under the curve (AUC) increases proportionally with dose. After administration of doses ≥ 60 mg, however, C_{max} increases less than proportionally with dose. The pharmacokinetic properties of tolvaptan are stereospecific, with a steady-state ratio of the S-(-) to the R-(+) enantiomer of about 3. The absolute bioavailability of SAMSCA is 56%. Peak concentrations of SAMSCA are observed between 2 and 4 hours post-dose. Food does not impact the bioavailability of SAMSCA. *In vitro* data indicate that SAMSCA is a substrate and inhibitor of P-gp. SAMSCA is highly plasma protein bound (99%) and distributed into an apparent volume of distribution of about 3 L/kg. SAMSCA is eliminated entirely by non-renal routes and mainly, if not exclusively, metabolized by CYP 3A. After oral dosing, clearance is about 4 mL/min/kg and the terminal phase half-life is about 12 hours. The accumulation factor of SAMSCA with the once-daily regimen is 1.3 and the trough concentrations amount to $\leq 16\%$ of the peak concentrations, suggesting a dominant half-life somewhat shorter than 12 hours. There is marked inter-subject variation in peak and average exposure to SAMSCA

with a percent coefficient of variation ranging between 30 and 60%.

In patients with hyponatremia of any origin the clearance of SAMSCA is reduced to about 2 mL/min/kg. Moderate or severe hepatic impairment or congestive heart failure decrease the clearance and increase the volume of distribution of SAMSCA, but the respective changes are not clinically relevant. Exposure and response to SAMSCA in subjects with creatinine clearance ranging between 79 and 10 mL/min and patients with normal renal function are not different.

In a study in patients with creatinine clearances ranging from 10-124 mL/min administered a single dose of 60 mg SAMSCA, AUC and C_{max} of plasma SAMSCA were less than doubled in patients with severe renal impairment relative to the controls. The peak increase in serum sodium was 5-6 mEq/L, regardless of renal function, but the onset and offset of SAMSCA's effect on serum sodium were slower in patients with severe renal impairment [see *Use in Special Populations* (8.7)].

12 NONCLINICAL TOXICOLOGY

12.1 Carcinogenesis, mutagenesis, impairment of fertility

Up to two years of oral administration of SAMSCA to male and female rats at doses up to 1000 mg/kg/day (162 times the maximum recommended human dose [MRHD] on a body surface area basis), to male mice at doses up to 60 mg/kg/day (5 times the MRHD) and to female mice at doses up to 100 mg/kg/day (8 times the MRHD) did not increase the incidence of tumors.

SAMSCA tested negative for genotoxicity in *in vitro* (bacterial reverse mutation assay and chromosomal aberration test in Chinese hamster lung fibroblast cells) and *in vivo* (rat micronucleus assay) test systems.

In a fertility study in which male and female rats were orally administered SAMSCA at 100, 300 or 1000 mg/kg/day, the highest dose level was associated with significantly fewer corpora lutea and implants than control.

12.2 Reproductive and developmental toxicology

In pregnant rats, oral administration of SAMSCA at 10, 100 and 1000 mg/kg/day during organogenesis was associated with a reduction in maternal body weight gain and food consumption at 100 and 1000 mg/kg/day, and reduced fetal weight and delayed ossification of fetuses at 1000 mg/kg/day (162 times the MRHD on a body surface area basis). Oral administration of SAMSCA at 100, 300 and 1000 mg/kg/day to pregnant rabbits during organogenesis was associated with reductions in maternal body weight gain and food consumption at all doses, and abortions at mid- and high-doses. At 1000 mg/kg/day (324 times the MRHD), increased incidences of embryo-fetal death, fetal microphthalmia, open eyelids, cleft palate, brachymelia and skeletal

malformations were observed. There are no adequate and well-controlled studies of SAMSCA in pregnant women. SAMSCA should be used in pregnancy only if the potential benefit justifies the risk to the fetus.

13 CLINICAL STUDIES

13.1 Hyponatremia

In two double-blind, placebo-controlled, multi-center studies (SALT-1 and SALT-2), a total of 424 patients with euvolemic or hypervolemic hyponatremia (serum sodium <135 mEq/L) resulting from a variety of underlying causes (heart failure, liver cirrhosis, syndrome of inappropriate antidiuretic hormone [SIADH] and others) were treated for 30 days with SAMSCA or placebo, then followed for an additional 7 days after withdrawal. Symptomatic patients, patients likely to require saline therapy during the course of therapy, patients with acute and transient hyponatremia associated with head trauma or postoperative state and patients with hyponatremia due to primary polydipsia, uncontrolled adrenal insufficiency or uncontrolled hypothyroidism were excluded. Patients were randomized to receive either placebo (N = 220) or SAMSCA (N = 223) at an initial oral dose of 15 mg once daily. The mean serum sodium concentration at study entry was 129 mEq/L. Fluid restriction was to be avoided if possible during the first 24 hours of therapy to avoid overly rapid correction of serum sodium, and during the first 24 hours of therapy, 87% of patients had no fluid restriction. Thereafter, patients could resume or initiate fluid restriction (defined as daily fluid intake of ≤1.0 liter/day) as clinically indicated.

The dose of SAMSCA could be increased at 24 hour intervals to 30 mg once daily, then to 60 mg once daily, until either the maximum dose of 60 mg or normonatremia (serum sodium >135 mEq/L) was reached. Serum sodium concentrations were determined at 8 hours after study drug initiation and daily up to 72 hours, within which time titration was typically completed. Treatment was maintained for 30 days with additional serum sodium assessments on Days 11, 18, 25 and 30. On the day of study discontinuation, all patients resumed previous therapies for hyponatremia and were reevaluated 7 days later. The primary endpoint for these studies was the average daily AUC for change in serum sodium from baseline to Day 4 and baseline to Day 30 in patients with a serum sodium less than 135 mEq/L. Compared to placebo, SAMSCA caused a statistically greater increase in serum sodium ($p < 0.0001$) during both periods in both studies (see [Table 3](#)). For patients with a serum sodium of <130 mEq/L or <125 mEq/L, the effects at Day 4 and Day 30 remained significant (see [Table 3](#)). This effect was also seen across all disease etiology subsets (e.g., CHF, cirrhosis, SIADH/other).

Table 3. Effects of Treatment with SAMSCA 15 mg/day to 60 mg/day

	SAMSCA 15 mg/day - 60 mg/day	Placebo	Estimated Effect (95% CI)
Subjects with Serum Sodium <135 mEq/L (ITT population)			
Change in average daily serum [Na ⁺] AUC baseline to Day 4 (mEq/L) Mean (SD) N	4.0 (2.8) 213	0.4 (2.4) 203	3.7 (3.3-4.2) <i>p</i> <0.0001
Change in average daily serum [Na ⁺] AUC baseline to Day 30 (mEq/L) Mean (SD) N	6.2 (4.0) 213	1.8 (3.7) 203	4.6 (3.9-5.2) <i>p</i> <0.0001
Percent of Patients Needing Fluid Restriction*	14% 30/215	25% 51/206	<i>p</i> =0.0017
Subgroup with Serum Sodium <130 mEq/L			
Change in average daily serum [Na ⁺] AUC baseline to Day 4 (mEq/L) Mean (SD) N	4.8 (3.0) 110	0.7 (2.5) 105	4.2 (3.5-5.0) <i>p</i> <0.0001
Change in average daily serum [Na ⁺] AUC baseline to Day 30 (mEq/L) Mean (SD) N	7.9 (4.1) 110	2.6 (4.2) 105	5.5 (4.4-6.5) <i>p</i> <0.0001
Percent of Patients Needing Fluid Restriction*	19% 21/110	36% 38/106	<i>p</i> <0.01
Subgroup with Serum Sodium <125 mEq/L			
Change in average daily serum [Na ⁺] AUC baseline to Day 4 (mEq/L) Mean (SD) N	5.7 (3.8) 26	1.0 (1.8) 30	5.3 (3.8-6.9) <i>p</i> <0.0001
Change in average daily serum [Na ⁺] AUC baseline to Day 30 (mEq/L) Mean (SD) N	10.0 (4.8) 26	4.1 (4.5) 30	5.7 (3.1-8.3) <i>p</i> <0.0001
Percent of Patients Needing Fluid Restriction*	35% 9/26	50% 15/30	<i>p</i> = 0.14

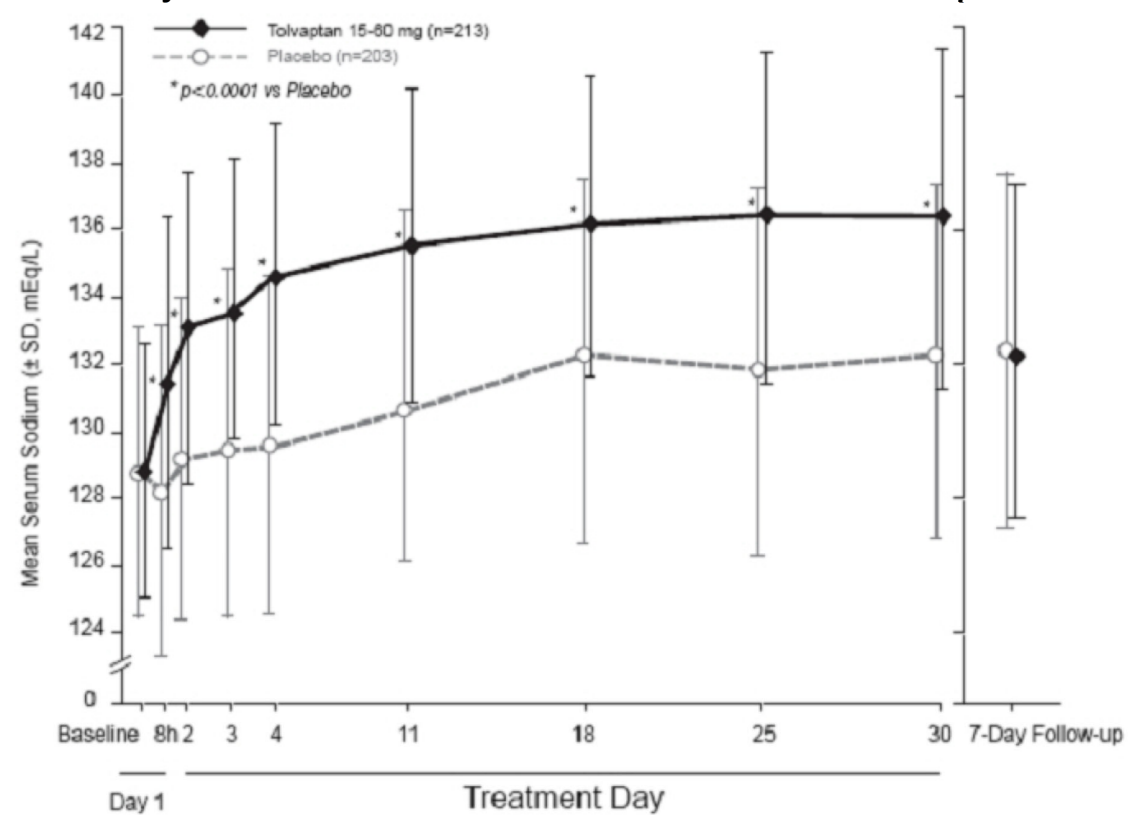
* Fluid Restriction defined as <1L/day at any time during treatment period.

In patients with hyponatremia (defined as <135 mEq/L), serum sodium concentration increased to a significantly greater degree in SAMSCA-treated patients compared to placebo-treated patients as early as 8 hours after the first dose, and the change was maintained for 30 days. The percentage of patients requiring fluid restriction (defined as ≤1 L/day at any time during the treatment period) was also significantly less (*p* =0.0017) in the SAMSCA-treated group (30/215, 14%) as compared with the placebo-treated group (51/206, 25%).

Figure 1 shows the change from baseline in serum sodium by visit in patients with serum sodium <135 mEq/L. Within 7 days of SAMSCA discontinuation, serum sodium

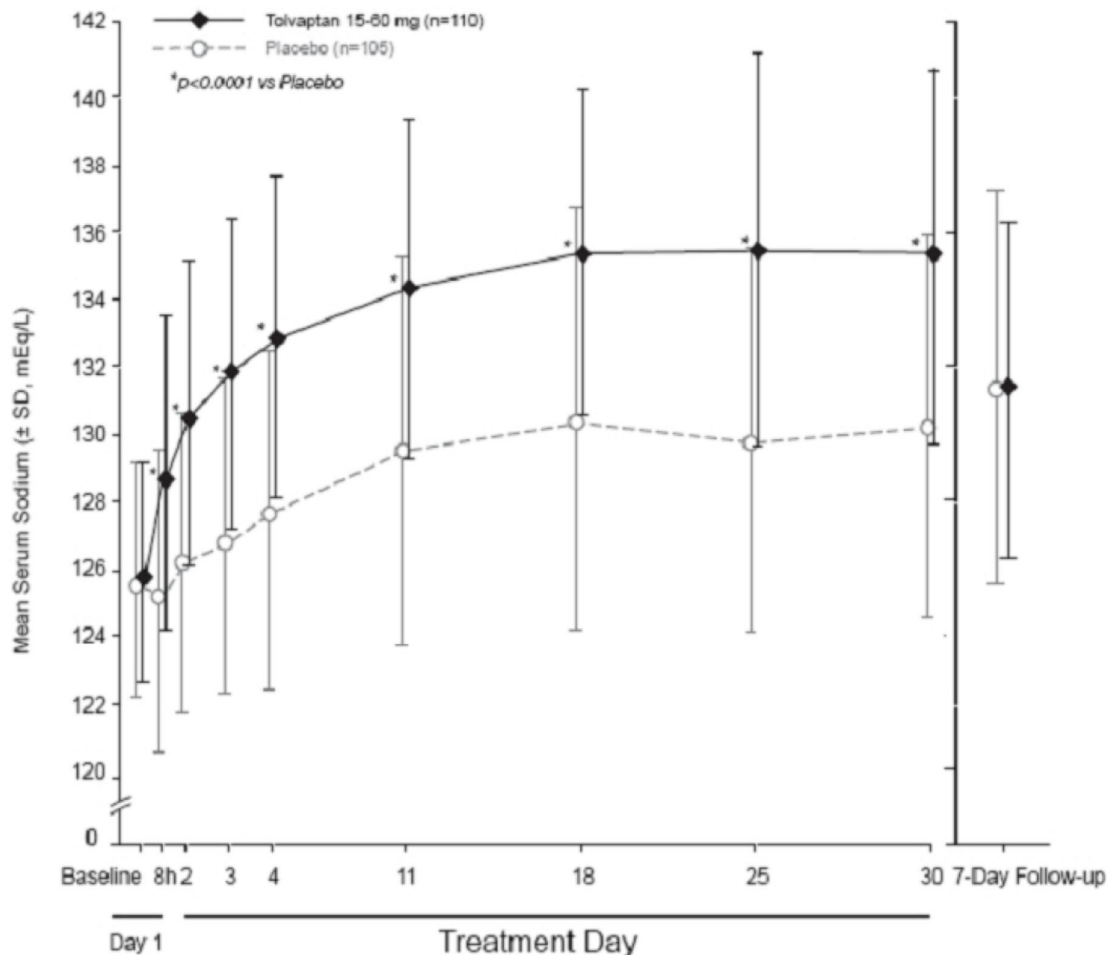
concentrations in SAMSCA-treated patients declined to levels similar to those of placebo-treated patients.

Figure 1: Pooled SALT Studies: Analysis of Mean Serum Sodium ($\pm$ SD, mEq/L) by Visit - Patients with Baseline Serum Sodium <135 mEq/L



*p-value <0.0001 for all visits during SAMSCA treatment compared to placebo

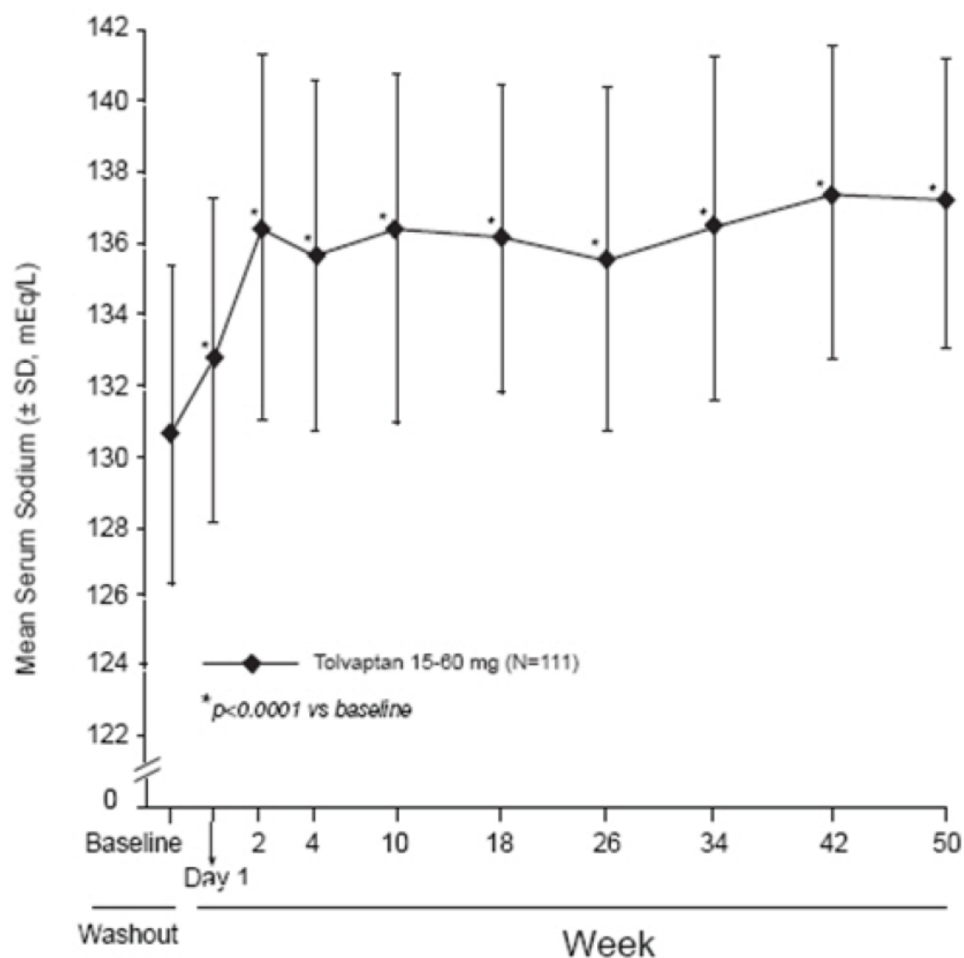
Figure 2: Pooled SALT Studies: Analysis of Mean Serum Sodium ($\pm$ SD, mEq/L) by Visit - Patients with Baseline Serum Sodium <130 mEq/L



*p-value <0.0001 for all visits during SAMSCA treatment compared to placebo

In the open-label study SALTWATER, 111 patients, 94 of them hyponatremic (serum sodium <135 mEq/L), previously on SAMSCA or placebo therapy were given SAMSCA as a titrated regimen (15 to 60 mg once daily) after having returned to standard care for at least 7 days. By this time, their baseline mean serum sodium concentration had fallen to between their original baseline and post-placebo therapy level. Upon initiation of therapy, average serum sodium concentrations increased to approximately the same levels as observed for those previously treated with SAMSCA, and were sustained for at least a year. [Figure 3](#) shows results from 111 patients enrolled in the SALTWATER Study.

Figure 3: SALTWATER: Analysis of Mean Serum Sodium ($\pm$ SD, mEq/L) by Visit



*p-value <0.0001 for all visits during SAMSCA treatment compared to baseline

13.2 Heart Failure

In a phase 3 double-blind, placebo-controlled study (EVEREST), 4133 patients with worsening heart failure were randomized to SAMSCA or placebo as an adjunct to standard of care. Long-term SAMSCA treatment (mean duration of treatment of 0.75 years) had no demonstrated effect, either favorable or unfavorable, on all-cause mortality [HR (95% CI): 0.98 (0.9, 1.1)] or the combined endpoint of CV mortality or subsequent hospitalization for worsening HF [HR (95% CI): 1.0 (0.9, 1.1)].

13.3 Adjunct treatment of volume overload in heart failure (Japanese Data)

The efficacy of oral administration of SAMSCA at 15 mg or placebo once daily for 7

days in congestive heart failure patients with volume overload despite the use of a conventional diuretic was evaluated in a double-blind, placebo-controlled phase 3 study. Changes in body weight from baseline at the end of treatment, the primary efficacy endpoint, were -1.54 ± 1.61 kg (mean $\pm$ SD) in the SAMSCA 15-mg group (baseline: 59.42 ± 12.30 kg, n=53) and -0.45 ± 0.93 kg in the placebo group (baseline: 55.68 ± 12.60 kg, n=57), showing that SAMSCA 15 mg significantly decreased body weight compared with placebo ($p < 0.0001$, t-test). A marked reduction in body weight was observed in the SAMSCA 15-mg group on Day 1, with subsequent decrease during the treatment period (Figure 4). Signs and symptoms associated with cardiac edema, including jugular venous distention, hepatomegaly, and lower limb edema, were also improved at the end of treatment (Table 4).

Figure 4: Change in Body Weight from Baseline in Patients with Cardiac Edema in a Placebo-controlled Double-blind Study

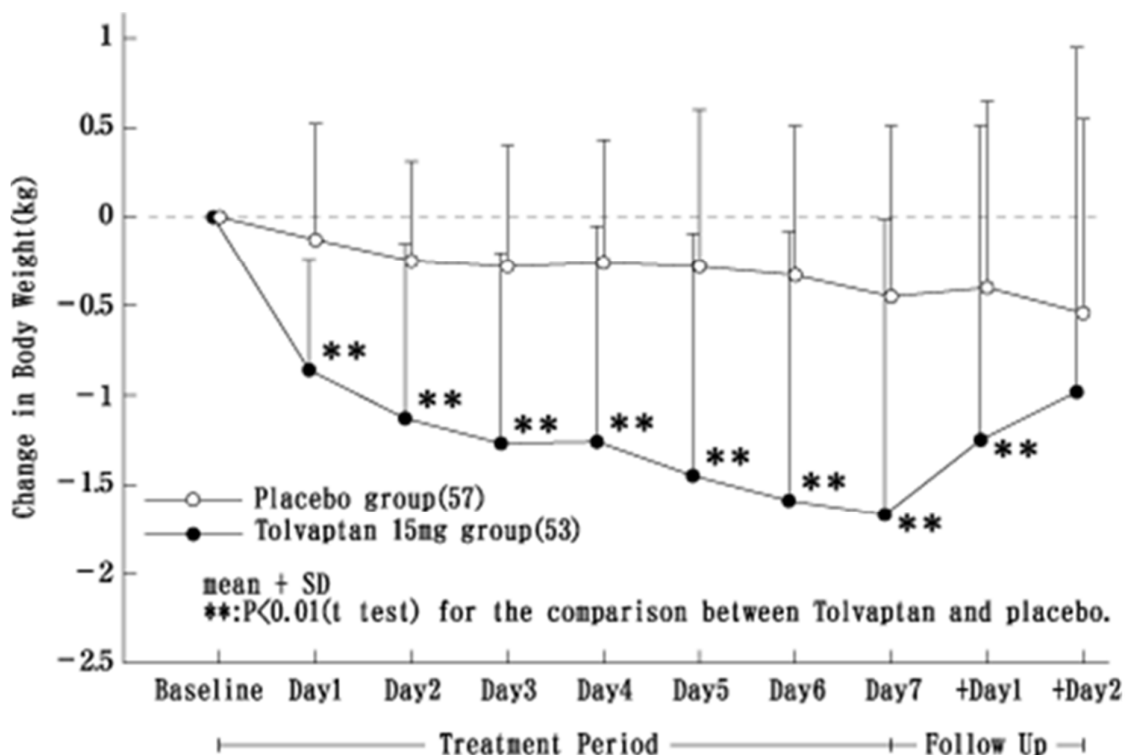


Table 4: Signs and Symptoms due to Cardiac Edema in a Placebo-controlled Double-blind Study

Signs and Symptoms Due to Cardiac Edema	SAMSCA 15 mg Group	Placebo Group
Change in jugular venous distention (cm) [n]	-2.03±2.81 [27]	-0.51±1.18 [19]
Change in hepatomegaly (cm) [n]	-1.07±0.89 [18]	-0.35±1.00 [17]
Improvement of lower limb edema (%) [n]	63.9 [23/36]	42.1 [16/38]

(Mean ± SD)

14 HOW SUPPLIED/STORAGE AND HANDLING

How Supplied

SAMSCA tablet is available in the following strength and package:

SAMSCA 15mg tablets are blue round tablets debossed with “SAMSCA”, “15” and cleavage line on one side. The cleavage line can be used to facilitate the halving of the tablet.

Blister of 10, 1 blister per box.

Storage and Handling

Store at or below 30°C.

Keep out of reach of children.

15 PATIENT COUNSELING INFORMATION

15.1 Concomitant Medication

Advise patients to inform their physician if they are taking or plan to take any prescription or over-the-counter drugs since there is a potential for interactions.

Strong and Moderate CYP 3A inhibitors and P-gp inhibitors

Advise patients to inform their physician if they use strong (e.g., ketoconazole, itraconazole, clarithromycin, telithromycin, nelfinavir, saquinavir, indinavir, ritonavir) or moderate CYP 3A inhibitors (e.g., aprepitant, erythromycin, diltiazem, verapamil, fluconazole) or P-gp inhibitors (e.g., cyclosporine) [see Dosage and Administration (2.3), Contraindications (4.5), Warnings and Precautions (5.6) and Drug Interactions

(7.1)].

15.2 Nursing

Advise patients not to breastfeed an infant if they are taking SAMSCA [see Use In Specific Populations (8.3)].

[MANUFACTURER]

Korea Otsuka Pharmaceutical Co., Ltd.
27, Jeyakgongdan 3-gil, Hyangnam-eup, Hwaseong-si, Gyeonggi-do, Republic of Korea

[PRODUCT REGISTRATION HOLDER]

Otsuka Pharmaceuticals (Malaysia) Sdn. Bhd.
Level 7, Oasis Wing, Brunsfield Oasis Tower 3
No. 2, Jalan PJU 1A/7A, Oasis Square
Oasis Damansara, 47301 Petaling Jaya
Selangor Darul Ehsan, Malaysia

Product under license from Otsuka Pharmaceutical Co., Ltd.

Mar 2022
Version 1